

Abhinav Kumar

Solution Architect | AWS Cloud Architecture | AI/LLM Platforms | Distributed Systems

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Architect Profile

Solution Architect **with 15+ years** of experience **designing, governing, and modernizing large-scale AWS-based enterprise platforms**, including serverless **AI/LLM architectures** and Retrieval-Augmented Generation (RAG) systems.

Own architecture decisions across application, integration, cloud, and platform domains, driving scalability, resiliency, performance, availability, security, and operational efficiency in **cloud-native architecture, distributed systems, microservices, event-driven processing and platform modernization**.

Led architecture initiatives supporting 310M+ subscribers and millions of billing transactions while enabling engineering teams to deliver reliable and scalable solutions.

Focused on building cost-efficient, scalable, and **responsible AI platforms** aligned with enterprise architecture and **governance standards**.

Skills

- **Architecture & Leadership:** Solution Architecture, Technical Strategy, Design Reviews, Technology Evaluation, Stakeholder Management
- **AWS & Cloud:** AWS Architecture, Serverless Architecture, Cloud-Native Design, Infrastructure as Code, Platform Engineering, CI/CD Architecture
- **Distributed Systems:** Microservices, Event-Driven Architecture, API Architecture, Integration Architecture, Domain Decomposition, Asynchronous Processing
- **Non-Functional Architecture:** Scalability, High Availability, Resiliency, Fault Tolerance, Performance Engineering, Cost Optimization

Technical Stack

- **AWS:** EC2, S3, Lambda, Aurora, RDS, ElastiCache, CloudFront, VPC, API Gateway, IAM, CloudWatch, SNS, SQS, EventBridge, Step Functions, RDS Proxy, EKS, ECS, Cognito, Secrets Manager, System Manager, AWS Budgets, Cost Explorer, Route 53, KMS
- **AI / GenAI:** Amazon Bedrock, Knowledge Base, Guardrails, Inference Profiles, Embedding Models, S3 Vectors (Vector Database), RAG, Tool Use (Function Calling), Agent Orchestration with AWS Strands Agents
- **Containers:** Kubernetes, Docker
- **Infrastructure as Code:** Terraform
- **CI/CD:** GitHub Actions

Professional Experience

Solution Architect – Telstra Global Services (Mar 2024 – Present)

- **Own architecture governance for AWS** platforms supporting 310M+ subscribers and high-volume billing transactions.
- Define **cloud-native target architectures** covering service boundaries, integration patterns, scalability, resiliency, observability, and governance.
- Architect **event-driven distributed systems on AWS** and lead **monolith-to-microservices transformation** to improve agility and reduce complexity.
- Design **asynchronous, resilient, multi-AZ architectures improving availability from ~99.2% to >99.9% and reducing cascading failures by ~40%.**
- Establish TMF-compliant **API standards supporting 30K–40K daily transactions** with rate limiting, fault tolerance, and resiliency controls.
- Conduct **architecture reviews, technical risk assessments, and technology evaluations** across multiple engineering teams.

Solution Architect – CRM Platforms – Netcracker Technology (Dec 2022 – Mar 2024)

- **Architected large-scale CRM and Telecom BSS platforms supporting high-volume customer and transaction workloads across multiple integrated systems**
- Defined service boundaries, domain ownership models, and integration architecture to improve scalability, maintainability, and deployment independence

Earlier Professional Experiences: (2011 – 2022)

- Solution Architect – Amdocs (Jan 2022 – Nov 2022)
- Subject Matter Expert – Amdocs (Mar 2015 – Dec 2021)
- Quality Analyst – Netcracker, (Jan 2014 – Feb 2015)
- Test Engineer - Tech Mahindra, (Jan 2011 – Dec 2013)

Portfolio Projects

- **Serverless AI / RAG Platform on AWS Bedrock**
[GitHub: https://github.com/abhinavcloud/Bedrock_AI](https://github.com/abhinavcloud/Bedrock_AI)
 - Designed and architected a cost-optimized **serverless RAG (AI/LLM)** platform on **AWS** using **Bedrock Knowledge Base, Titan v2 embeddings, and S3 Vectors**, eliminating OpenSearch dependency and **reducing baseline cost from ~\$350/month to <\$0.01/month**.
 - Architected end-to-end AI pipeline: **S3 → Bedrock Knowledge Base → embeddings → vector index → RetrieveAndGenerate → grounded LLM responses** with **guardrails**.
 - Implemented **Bedrock Guardrails** (content filters, denied topics, PII redaction) ensuring safe and compliant AI responses in production scenarios.
 - Designed **Lambda-based ingestion** trigger (StartIngestionJob), leveraging **Bedrock** for **async processing** and eliminating compute bottlenecks.
 - Implemented **progressive GenAI** capabilities: **Converse API, knowledge base retrieval, multi-turn conversations, tool use (function calling)**, and **agent orchestration** using **AWS Strands Framework**.

- **Online Ticketing System**

Website: mytickets.click

GitHub: [https://github.com/abhinavcloud/Online Ticketing System](https://github.com/abhinavcloud/Online-Ticketing-System)

- Designed and implemented a Terraform-driven AWS ticketing platform using modular Network, Database, Cache, Compute, Authentication, API Gateway, and Notification layers with private VPC wiring and service-to-service IAM integration.
- Built a **production-grade serverless booking architecture** using **API Gateway, Lambda, Cognito, Aurora PostgreSQL Serverless v2, RDS Proxy, ElastiCache Serverless (Valkey), KMS, and SNS** to demonstrate secure, scalable, event-driven backend design.
- Designed a **high-contention booking flow** with **queue-based admission control, cache-backed seat locking, and all-or-nothing multi-seat reservation logic** to prevent partial allocation and overselling under concurrency.
- Implemented **Cognito-based authentication with Google OAuth** and integrated protected booking APIs with API Gateway authorizers while keeping browse APIs publicly accessible.

- **Production-grade, serverless URL Shortener**

Website App: lshort.in

GitHub: <https://github.com/abhinavcloud/shortURL>

- Designed and implemented a production-grade serverless architecture **API Gateway, Lambda, DynamoDB, Cognito, CloudFront, and S3** demonstrating scalability, security, and global content delivery patterns.
- Implemented **Cognito-based authentication with Google OAuth**.
- Automated CI/CD using **Terraform** and **GitHub Actions**.

- **Multi Environment AWS Production Ready Three Tier Architecture**

GitHub: <https://github.com/abhinavcloud/Three-Tier-AWS-Architecture-Terraform>

- Designed reusable multi-environment cloud architecture patterns supporting scalable and secure deployments
- Defined modular Terraform architecture enabling consistency and environment isolation

Certifications

Credly: <https://www.credly.com/users/abhinav-kumar.9c24f238/badges/credly>

- **Cloud:** AWS Solutions Architect – Associate, AWS Developer – Associate, AWS SysOps Administrator – Associate
- **Infrastructure as Coded (Iac):** Hashicorp Terraform Associate
- **Microservices and Container Orchestration:** Certified Kubernetes Administrator (CKA), Kubernetes and Cloud Native Associate (KCNA)
- **CI/CD:** Microsoft GitHub Actions Associate
- **AI:** AWS AI Practitioner

Education:

Bachelor's in technology (Graduation in Electronics and Computers), ICFAI University, Dehradun, 2006-2010.